

GOSFORD HIGH SCHOOL



2009

ASSESSMENT TASK 3

MATHEMATICS EXTENSION 1

Time allowed: 70 minutes (plus 5 minutes reading time)

Name: _____

General Instructions:

- Write using a black or blue pen.
- Show all necessary working.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.

TOTAL MARKS: 50

- Attempt all Sections 1 – 4
- Sections are **NOT** of equal value.
- Start each **Section** on a new sheet of paper.

MARKERS USE ONLY

Section 1:		Total
I. Integration		/14
II. Series		
Section 2:		
I. Parametric Representation		/12
II. Mathematical Induction		
Section 3:		
Inverse Functions		/10
Section 4:		
Inverse Trigonometric Functions		/14
TOTAL		/50

SECTION 1 (14 Marks) Use a new sheet of paper.

Marks

I. INTEGRATION

a. Find $\int \cos^2 3x \, dx$

2

b. Evaluate, in exact form:

$$\int_0^1 x(1-2x)^3 \, dx$$

3

Using the substitution $u = 1 - 2x$

c. By using $u = 1 + e^x$, show that

$$\int_0^1 \frac{e^{3x}}{1+e^x} \, dx = \frac{(e-1)^2}{2} + \ln\left(\frac{e+1}{2}\right)$$

3

Section 1 continued

II. SERIES

- a. If a , b and c are consecutive terms in an arithmetic series, show that e^a, e^b and e^c are consecutive terms in a geometric series.

1

- b. Find the sum of the first 15 terms of the series:

2

$$\log_{10} 3 + \log_{10} 27 + \log_{10} 243 + \dots$$

(Answer in exact form)

- c. For the infinite geometric series:

$$2 + \frac{4}{x+5} + \frac{8}{(x+5)^2} + \dots$$

- i. Find the limiting sum when $x = 5$

1

- ii. For what values of x will the above series have a limiting sum?

2

SECTION 2 (12 Marks) Use a new sheet of paper.

Marks

I. PARAMETRIC REPRESENTATION

- a. Given the parametric equations

$$x = 8t \quad \text{and} \quad y = 4t^2$$

1

Eliminate t to find the Cartesian equation of the parabola

- b. Given the parabola $x^2 = 4y$

- i. Show that the equation of the tangent at the point $P(2t, t^2)$ is given by $y = tx - t^2$

2

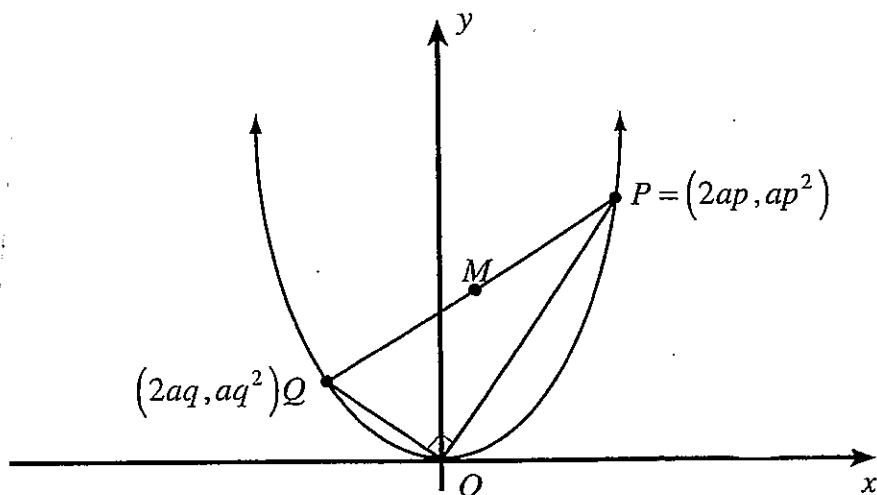
- ii. Find the equation of the two tangents to the parabola that could be drawn from the point $(1, -6)$

2

- iii. Find the co-ordinates of the two points of contact of these tangents.

1

- c. $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. PQ subtends a right angle at the vertex O of the parabola.



- i. Prove that $pq = -4$
- ii. Find the equation of the locus of the midpoint M of the chord PQ.

1

2

Section 2 Continued

Marks**II. MATHEMATICAL INDUCTION**

- a. Show by mathematical induction that for all positive integers $n \geq 1$

$$\cos(x + n\pi) = (-1)^n \cos x$$

3

SECTION 3 (10 Marks) Use a new sheet of paper.

Marks

INVERSE FUNCTIONS

- a. Consider the function $y = x^2 - 2$
- Sketch $y = f(x)$ showing x and y intercepts. 1
 - Find the largest positive domain for which $y = f(x)$ has an inverse function $y = f^{-1}(x)$ 1
 - State the domain of $y = f^{-1}(x)$ 1
 - Sketch the graph of $y = f^{-1}(x)$ in this domain 1
- b. Given the function of $g(x) = 2 + \frac{4}{x-3}$ for $x > 3$
- Sketch the function $y = g(x)$ clearly indicating all intercepts and asymptotes. 2
 - Find an expression for the inverse function $y = g^{-1}(x)$ in terms of x 2
 - Find the x co-ordinate, correct to 2 decimal places, of the point of intersection of $y = g(x)$ and $y = g^{-1}(x)$ 2

SECTION 4 (14 Marks) Use a new sheet of paper.

Marks

INVERSE TRIGONOMETRIC FUNCTIONS

- a. Given that $f(x) = 3 \cos^{-1} \frac{x}{2}$, evaluate $f(1)$ in exact form 1
- b. Find $\frac{dy}{dx}$ if $y = (2x+1)^3 \tan^{-1} x$ 2
- c. Find $\int \frac{dx}{\sqrt{1-49x^2}}$ 2
- d. The portion of the curve $y = \frac{1}{\sqrt{4+x^2}}$ between $x=0$ and $x=2$ is rotated about the x axis. Find the exact volume of the solid of revolution formed. 2
- e. Let $h(x) = \sin^{-1}(x+3)$
- i. State the domain and range of $h(x)$ 2
- ii. Find the equation of the normal to the curve $y=h(x)$ at the point where $x = -3$ 2
- iii. Sketch the graph of $y=h(x)$ 1
- f. Evaluate $\sin \left[\cos^{-1} \left(\frac{4}{5} \right) + \tan^{-1} \left(-\frac{4}{3} \right) \right]$ 2

End of Test

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SECTION 1

1. Integration

$$\begin{aligned} \text{a. } \int \cos^2 3x \, dx &= \frac{1}{2} \int (1 + \cos 6x) \, dx \\ &= \frac{1}{2} \left(x + \frac{\sin 6x}{6} \right) + C \end{aligned}$$

$$\begin{aligned} \text{b. } \int_0^1 x(1-2x)^3 \, dx & \quad \left\{ \begin{array}{l} u = 1-2x \\ \frac{du}{dx} = -2 \\ \therefore dx = -\frac{du}{2} \end{array} \right. \\ &= \int_1^{-1} \frac{1-u}{2} (u)^3 \cdot \frac{-du}{2} \\ &= \frac{1}{4} \int_{-1}^1 (1-u)u^3 \, du \quad \left\{ \begin{array}{l} \text{Also } x = \frac{1-u}{2} \\ x=1 \quad u=-1 \\ x=0 \quad u=1 \end{array} \right. \\ &= \frac{1}{4} \int_{-1}^1 u^3 - u^4 \, du \\ &= \frac{1}{4} \left[\frac{u^4}{4} - \frac{u^5}{5} \right]_{-1}^1 \\ &= \frac{1}{4} \left[\left(\frac{1}{4} - \frac{1}{5} \right) - \left(\frac{1}{4} - \frac{1}{5} \right) \right] = -\frac{1}{20} \end{aligned}$$

$$\begin{aligned} \text{c. } \int_0^1 \frac{e^{3x}}{1+e^x} \, dx & \quad \left\{ \begin{array}{l} u = 1+e^x \\ \frac{du}{dx} = e^x \\ \frac{du}{e^x} = dx \end{array} \right. \\ &= \int_2^{1+e} \frac{e^{2x}}{u} \, du \quad \left\{ \begin{array}{l} x=1 \quad u=1+e \\ x=0 \quad u=2 \end{array} \right. \\ & \quad (\text{Now } e^x = u-1 \\ & \quad \therefore e^{2x} = (u-1)^2) \end{aligned}$$

$$\begin{aligned} &= \int_2^{1+e} \frac{(u-1)^2}{u} \, du \\ &= \int_2^{1+e} \frac{u^2 - 2u + 1}{u} \, du \\ &= \int_2^{1+e} \left(u - 2 + \frac{1}{u} \right) \, du \\ &= \left[\frac{u^2}{2} - 2u + \ln u \right]_2^{1+e} \\ &= \left[\left(\frac{(1+e)^2}{2} - 2(1+e) + \ln(1+e) \right) - \left(\frac{2^2}{2} - 4 + \ln 2 \right) \right] \\ &= \frac{1 + 2e + e^2 - 4 - 4e}{2} + \ln\left(\frac{1+e}{2}\right) + 2 \\ &= \frac{(e-1)^2}{2} + \ln(1+e) + 2 \end{aligned}$$

11. Series

$$\text{a. } b-a = c-b$$

$$\therefore e^{b-a} = e^{c-b}$$

$$\frac{e^b}{e^a} = \frac{e^c}{e^b}$$

$\therefore e^a, e^b, e^c$ form a G.P.

$$\text{b. } \log_{10} 3 + \log_{10} 3^3 + \log_{10} 3^5 + \dots$$

$$= \log_{10} 3 + 3 \log_{10} 3 + 5 \log_{10} 3 + \dots$$

$\therefore$ Arithmetic with

$$a = \log_{10} 3, d = 2 \log_{10} 3$$

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\begin{aligned} S_{15} &= \frac{15}{2} [2 \log_{10} 3 + 14(2 \log_{10} 3)] \\ &= 225 \log_{10} 3. \end{aligned}$$

$$\text{c. } 2 + \frac{4}{x+5} + \frac{8}{(x+5)^2} + \dots$$

(1) when $x \neq -5$

$$2 + \frac{4}{10} + \frac{8}{100} + \dots$$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{2}{1-\frac{2}{10}} \\ &= \frac{5}{2} \end{aligned}$$

(11) The series will have a S_{∞} when $-1 < r < 1$

$$\text{or } |r| < 1$$

$$\text{i.e. } \left| \frac{2}{x+5} \right| < 1 \quad (V: x \neq -5)$$

Consider

$$\left| \frac{2}{x+5} \right| = 1$$

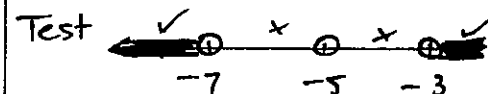
$$\text{i.e. } \frac{2}{x+5} = 1 \quad \text{or} \quad \frac{2}{x+5} = -1$$

$$2 = x+5$$

$$2 = -x-5$$

$$-3 = x$$

$$x = -7$$

Test 

$$\text{i.e. } x < -7 \quad \text{or} \quad x > -3$$

SECTION 2

1. Parametric Representation

$$\begin{aligned} \text{a. } x &= 8t \quad - (1) \\ y &= 4t^2 \quad - (2) \end{aligned}$$

$$\text{From (1) } t = \frac{x}{8}$$

$$\text{Sub in (2) } y = \frac{x^2}{16}$$

$$\therefore x^2 = 16y$$

$$\text{b. (i) } y = \frac{x^2}{4}$$

$$\therefore \frac{dy}{dx} = \frac{x}{2}$$

$$\text{when } x = 2t, \quad m = t$$

$\therefore$ eqn of tangent:

$$y - t^2 = t(x - 2t)$$

$$y - t^2 = tx - 2t^2$$

$$y = tx - t^2 \text{ as req.}$$

$$\text{(ii) } x = 1, y = -6$$

$$\therefore -6 = t - t^2$$

$$t^2 - t - 6 = 0$$

$$(t-3)(t+2) = 0$$

$\therefore$ 2 tangents are:

$$y = 3x - 9 \text{ and}$$

$$y = -2x - 4$$

$$\begin{aligned} \text{(iii) } t = 3 \quad P &= (6, 9) \\ t = -2 \quad P &= (-4, 4) \end{aligned}$$

$$\begin{aligned} \text{c. (i) } M_{OP} &= \frac{ap^2 - 0}{2ap - 0} \\ &= \frac{ap^2}{2ap} \\ &= \frac{p}{2} \end{aligned}$$

$$\text{Similarly } M_{OQ} = \frac{q}{2}$$

If $PO \perp QO$ then

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$\therefore pq = -4 \text{ as req.}$$

$$\begin{aligned} \text{(ii) } M &= \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right) \\ &= \left(a(p+q), a \left(\frac{p^2 + q^2}{2} \right) \right) \end{aligned}$$

$$\therefore x = a(p+q) \quad - (1)$$

$$y = a \left(\frac{p^2 + q^2}{2} \right) \quad - (2)$$

$$(2) \Rightarrow y = \frac{a}{2} \left[(p+q)^2 - 2pq \right]$$

Since from c(i) $pq = -4$
and from (1) $p+q = \frac{x}{a}$

$$(2) \Rightarrow y = \frac{a}{2} \left[\left(\frac{x}{a} \right)^2 + 8 \right]$$

$$y = \frac{x^2}{2a} + 4a$$

$$2a(y - 4a) = x^2$$

$$\therefore x^2 = 2ay - 8a^2$$

II. Mathematical Induction

Step 1. To prove true for $n=1$

$$\text{LHS} = \cos(x+\pi) = -\cos x$$

$$\text{RHS} = (-1)^1 \cos x = -\cos x$$

$$\therefore \text{LHS} = \text{RHS}$$

and statement is true for $n=1$.

Step 2. Assume it is true for $n=k$

$$\text{i.e. } \cos(x+k\pi) = (-1)^k \cos x$$

Step 3. To prove true for $n=k+1$

$$\begin{aligned} \text{LHS: } \cos(x + (k+1)\pi) &= \cos((x+k\pi) + \pi) \\ &= \cos(x+k\pi)\cos\pi - \sin(x+k\pi)\sin\pi \\ &= (-1)^k \cos x \cdot (-1) - \sin(x+k\pi) \cdot 0 \\ &= (-1)^{k+1} \cos x \end{aligned}$$

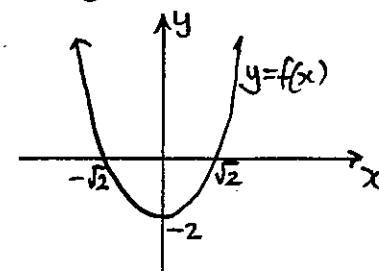
$\therefore$ result holds for $n=k+1$ if true for $n=k$.

$\therefore$ proved by induction.

SECTION 3

INVERSE FUNCTIONS

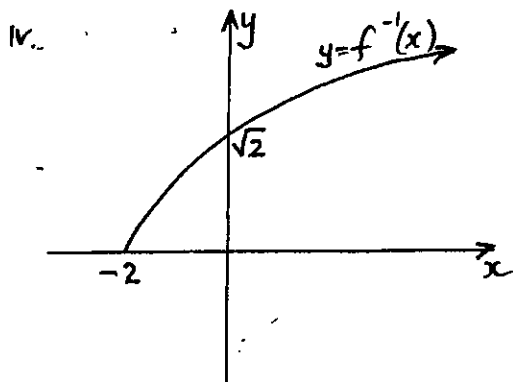
$$\text{a. (i) } y = x^2 - 2$$



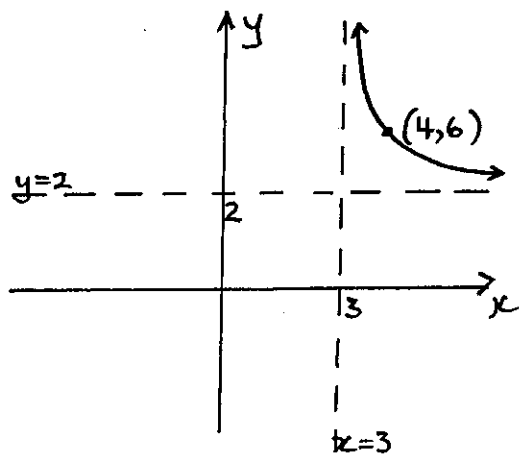
$$\text{(ii) } x \geq 0$$

(iii) Domain of $y = f^{-1}(x)$ is the range of $y = f(x)$.

$\therefore$ Domain is $x \geq -2$



b.(i) $g(x) = 2 + \frac{4}{x-3}$



(ii) $y = 2 + \frac{4}{x-3}$

$g^{-1}(x) \Rightarrow x = 2 + \frac{4}{y-3}$

$x-2 = \frac{4}{y-3}$

$(x-2)(y-3) = 4$

$xy - 3x - 2y + 6 = 4$

$y(x-2) = 3x-2$

$y = \frac{3x-2}{x-2}$

(iii) To find the point of intersection

solve $y = x$ — (1)

$y = \frac{3x-2}{x-2}$ — (2)

Sub (1) in (2)

$x = \frac{3x-2}{x-2}$

$x^2 - 2x = 3x - 2$

$x^2 - 5x + 2 = 0$

$x = \frac{+5 \pm \sqrt{25-8}}{2}$

$= \frac{5 \pm \sqrt{17}}{2}$

$= \frac{5+\sqrt{17}}{2}, \frac{5-\sqrt{17}}{2}$ not in req. domain

$\therefore x = 4.56$ (2 d.p)

SECTION 4

Inverse Trig. Functions

a. $f(x) = 3 \cos^{-1} \frac{x}{2}$

$f(1) = 3 \cos^{-1} \frac{1}{2}$

$= 3 \cdot \frac{\pi}{3}$

$= \pi$

b. $y = (2x+1)^3 \tan^{-1} x$

$\frac{dy}{dx} = \tan^{-1} x \cdot 3(2x+1)^2 \cdot 2$
 $+ (2x+1)^3 \cdot \frac{1}{1+x^2}$

$= 6(2x+1)^2 \tan^{-1} x + \frac{(2x+1)^3}{1+x^2}$

c. $\int \frac{dx}{\sqrt{1-49x^2}}$

$= \int \frac{dx}{\sqrt{49(\frac{1}{49} - x^2)}}$

$= \frac{1}{7} \sin^{-1} 7x + C$

d. $V = \pi \int_0^2 \left(\frac{1}{\sqrt{4+x^2}} \right)^2 dx$ 3.

$= \pi \int_0^2 \frac{1}{4+x^2} dx$

$= \pi \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$

$= \frac{\pi}{2} \left[\tan^{-1} 1 - \tan^{-1} 0 \right]$

$= \frac{\pi}{2} \left[\frac{\pi}{4} - 0 \right]$

$= \frac{\pi^2}{8} \text{ units}^3$

e.(i) $h(x) = \sin^{-1}(x+3)$

Domain: $-1 \leq x+3 \leq 1$
 $-4 \leq x \leq -2$

Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(ii) $y = \sin^{-1}(x+3)$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x+3)^2}} \times 1$

$$= \frac{1}{\sqrt{1-(x+3)^2}}$$

when $x = -3$

$$\frac{dy}{dx} = 1$$

$$\therefore m_{\text{normal}} = -1$$

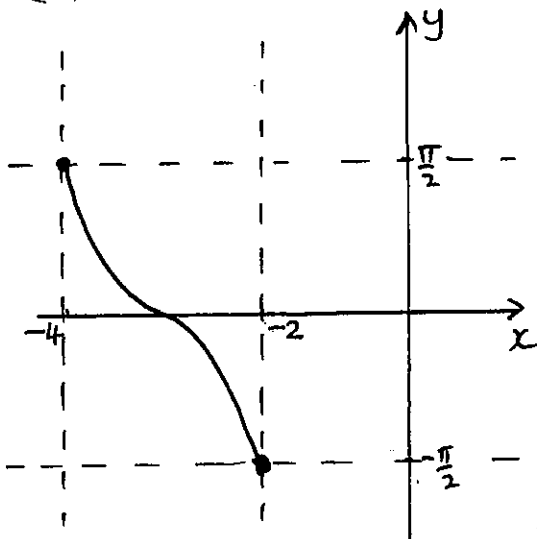
and equation of normal is

$$y - 0 = -1(x + 3)$$

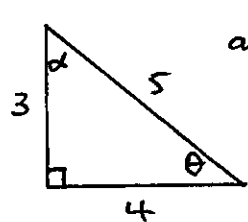
$$y = -x - 3$$

$$x + y + 3 = 0$$

(iii)



(f) Using:



$$\text{and } \tan^{-1}(-x) = -\tan^{-1}x$$

$$\cos \theta = \frac{4}{5} \quad \tan \alpha = \frac{4}{3}$$

$$\therefore \cos^{-1}\left(\frac{4}{5}\right) = \theta \quad \tan^{-1}\frac{4}{3} = \alpha$$

$$\therefore \tan^{-1}\left(-\frac{4}{3}\right) = -\alpha$$

$$\therefore \sin\left[\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(-\frac{4}{3}\right)\right]$$

$$= \sin(\theta - \alpha)$$

$$= \sin \theta \cos \alpha - \sin \alpha \cos \theta$$

$$= \frac{3}{5} \cdot \frac{3}{5} - \frac{4}{5} \cdot \frac{4}{5}$$

$$= \frac{9}{25} - \frac{16}{25}$$

$$= -\frac{7}{25}$$